

Vaibhav Kumar Mahawar

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Portfolio | Ajmer, Rajasthan, India

Education

Dr. B. R. Ambedkar National Institute of Technology Jalandhar <i>Master of Technology in Machine Intelligence and Automation</i> CGPA: 8.48/10	Jalandhar, Punjab Aug. 2024 – Jun. 2026
Poornima Institute of Engineering and Technology <i>Bachelor of Technology in Computer Science and Engineering</i> CGPA: 8.54/10	Jaipur, Rajasthan Jul. 2019 – Jun. 2023

Experience

Software Development Intern <i>Techienest Pvt. Ltd.</i>	Jul. 2022 – Sep. 2022 Jaipur, Rajasthan
- Developed a React Native mobile application serving 500+ students by integrating REST APIs, progress tracking, and interactive learning modules while improving application usability. - Collaborated with cross-functional teams to develop, test, and optimize scalable mobile application features while integrating REST APIs and improving application performance and maintainability.	

Projects

RudraAI– Intelligent Software Engineering Platform <i>Node.js, Typescript, Python, Redis, Neo4j, Docker</i>	2026
- Developed a scalable software engineering platform using FastAPI, Node.js, Redis, Neo4j, and Docker to automate repository analysis, code generation, debugging, documentation, and developer workflows. - Designed scalable backend services with FastAPI, Redis, and Neo4j while implementing memory-enabled reasoning, tool execution, and intelligent agent collaboration for software engineering workflows.	
Career Twin– Full-Stack Career Platform <i>React, Typescript, AWS Bedrock, LangGraph, Vector Search</i>	2026
- Developed a full-stack career platform using React, TypeScript, FastAPI, and AWS Bedrock for resume analysis, job matching, and interview preparation. - Built scalable backend APIs integrating vector search, semantic retrieval, and document processing to deliver personalized career recommendations.	
Drone Surveillance Analytics Platform <i>Python, YOLOv8, OpenCV, LangChain, LLMs, ANFIS, RAG</i>	2026
- Developed a modular surveillance analytics platform integrating computer vision inference pipelines with backend processing for drone detection and threat analysis. - Implemented backend services for surveillance summarization, contextual analytics, and natural language interaction over detection results.	

Research Experience

Real-Time Drone Detection and Classification System <i>Python, YOLOv8, OpenCV, ANFIS, Fuzzy Logic</i>	2026
Developed a real-time YOLOv8-based drone detection framework with ANFIS-enhanced classification, achieving over 91% detection accuracy while reducing bird–drone misclassification by 52%.	
Generative AI and Agentic Systems Research <i>LLMs, LangChain, LangGraph, MCP, RAG, AI Agents</i>	Ongoing
Exploring autonomous AI agents, Retrieval-Augmented Generation (RAG), Model Context Protocol (MCP), long-context reasoning, and multi-agent orchestration for enterprise-scale AI applications.	

Technical Skills

Programming Languages: Python, Java, C++, JavaScript, TypeScript, SQL
AI & Machine Learning: Scikit-learn, OpenCV, YOLO, Computer Vision, CNNs, Transformers, Vision Transformers, OpenAI, Gemini, LangChain, LangGraph, RAG, MCP, AI Agents, Embeddings, Vector Search
Backend Development: FastAPI, REST APIs, Node.js, Express.js
Frontend Development: React.js, React Native, HTML, CSS
Databases: Neo4j, Redis, PostgreSQL, MongoDB, SQLite, FAISS, Weaviate
Cloud & Devops AWS, Docker, Kubernetes, Git, GitHub, GitHub Actions, Linux, Azure
Tools: VS Code, Eclipse, Jupyter Notebook, Postman, Linux

Certifications & Achievements

Research Publications Published research papers in international journals and conferences on Computer Vision, Artificial Intelligence, Intelligent Surveillance, Photovoltaic Systems, and Renewable Energy Optimization.
Microsoft Certified: Azure AI Fundamentals (AI-900) Certified in Azure AI services, Machine Learning fundamentals, Responsible AI, and AI solution development.
GATE Qualified Qualified GATE Computer Science (2024 & 2026) and GATE Data Science & Artificial Intelligence (DA 2026).